

A STUDY OF VACCINAL IMMUNITY IN TISSUE CULTURES

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In the preceding paper (1) a method of obtaining in tissue cultures visible evidences of infection with vaccine virus was described. The method is applicable to the study of certain phenomena related to vaccinal immunity, and the present communication deals with information secured in this manner.

For the investigation of various immunological problems, tissue cultures or tissue culture methods have been employed by Carrel and Ingebrigtsen (2), Przygode (3, 4), Foot (5), Hadda and Rosenthal (6), Lambert and Hanes (7), Bloom (8), Schwartzman (9, 10), Meyer and Loewenthal (11), Loewenthal and Micseh (12), and others. Such methods have also been used in experimental work on virus immunity. Steinhardt and Lambert (13) attempted to grow vaccine virus in cultures of normal corneal epithelium embedded in antivaccinal plasma. When these cultures were emulsified and inoculated on the skin of a rabbit, no virus was demonstrable. Later, Harde (Steinhardt) (14) reported the results of additional work. She found that cultures of normal or immune corneas in antivaccinal plasma did not support the growth or survival of vaccine virus, while preparations of immune corneas in normal plasma were less "lytic" for the virus. Recently, Nye and Parker (15) attempted to cultivate vaccine virus in normal tissues embedded in normal plasma to which serum from a hyperimmunized calf had been added. In these cultures they were unable to demonstrate virus, while in their control preparations virus was shown to be present.

Fischer's (16) results with the virus of Rous' sarcoma are at variance with those reported above by workers dealing with vaccine virus. He found that serum from immune fowls neutralized active tumor filtrates, but had no deleterious effects on sarcoma cells in tissue cultures, for, when cells from such cultures were injected into susceptible fowls, typical tumors developed. In view of these findings, he concluded that the antibodies in the immune sera were unable to injure virus bound to or situated within cells.

Todd (17) and Andrewes (18) have shown that certain antiviral sera, which inhibit the activity of viruses, are not necessarily "virucidal." From inactive mixtures or from neutral mixtures of immune serum and virus they have been able, under proper conditions, to recover active fowl plague virus and vaccine virus.

Furthermore, Andrewes (19) has reported that, if vaccine virus was injected into the skin 5 minutes before an overneutralizing dose of antivaccinal serum was administered in and around the site of inoculation, the activity of the infectious agent was not completely suppressed; whereas, if the serum was injected first, or if a mixture of virus and serum was injected, no vaccinal lesions developed. From these facts he is prepared to entertain the idea that the protective antibodies in the serum act by rendering susceptible cells insusceptible rather than by killing the virus.

Rivers and Pearce (20) found that it was possible to recover Virus III and vaccine virus from a transplantable rabbit neoplasm for at least 54 and 64 days respectively after a mixture of tumor cells and viruses had been inoculated intratesticularly. At the time when the viruses were recovered, the hosts had long since become refractory to reinoculation from without, and their sera had been "virucidal" for many weeks. Douglas, Smith, and Price (21) have obtained vaccine virus from the brains of recovered animals 41 days after inoculation. By means of cataphoresis, Olitsky and Long (22) demonstrated vaccine virus in the testicles of rabbits inoculated in the skin many months previously. Cole and Kuttner (23) were able at all times to obtain from immune animals the "salivary-gland virus" of guinea pigs, and it appears that a pig once infected continues to harbor the virus indefinitely in spite of a refractory state to reinfection from without.

The above review of the literature concerning virus immunity is not exhaustive. It is given merely to point out some of the discordant results obtained by means of tissue cultures, and to indicate a few phenomena, observed in animals infected with viruses, that are difficult or impossible of analysis in a living host. The experiments described below, in which the visible evidences of vaccinal infection in tissue cultures, inclusion bodies, that is to say, guided or aided the workers, show how certain problems in the virus field may be attacked.

Methods and Materials

Virus.—Levaditi's neurovaccine virus prepared in the testicles of rabbits was employed in all experiments. It was found advisable to use a freshly prepared virus emulsion to which no preservative had been added.

Tissues.—Rabbit corneas prepared and handled as described in the preceding paper (1) constituted the tissues investigated. Normal corneas were obtained from stock animals. Immune corneas were secured from rabbits that had recovered from severe vaccinal infections in the skin or testicles. Neutralizing properties in the blood and a refractory state in the skin develop rapidly after a dermal infection with vaccine virus. An immune state of the cornea, however, develops very slowly under such conditions, and at times a completely refractory

condition requires several months for its appearance. In this work, no cornea was considered immune until it had been tested *in vivo* with an extremely potent virus and found refractory, or until it had recovered from the vaccinal keratoconjunctivitis induced by the test inoculation. Furthermore, it is not expedient to test one eye and then use the other, because both eyes do not always simultaneously attain an absolutely refractory state.

Plasma and Serum.—For each experiment fresh normal plasma or serum was collected from stock rabbits. Immune plasma or serum was obtained from rabbits that had recovered, 2 to 4 weeks previously, from severe vaccinal infections. The amount of immune plasma or serum used at any time in the experiments was known to be more than sufficient for the neutralization of the virus employed.

Tissue Extract.—Tissue extract was always secured from the spleens of rabbits that had supplied the normal corneas and normal plasma.

Cultures.—The methods of inoculating the tissues, and of preparing, examining, and testing the cultures were similar to those described in the preceding paper (1).

EXPERIMENTAL

Numerous experiments have been performed, in which various combinations of normal and immune corneas with normal and immune sera, or with normal and immune plasmas, were tested for their reactions to, or effects on vaccine virus. For instance, bits of normal corneal tissue were immersed in immune serum for 2 hours at 37°C., washed in Ringer's solution, placed in an emulsion containing vaccine virus for 2 hours, again washed in Ringer's solution, and finally cultivated in normal plasma and spleen extract. The reverse experiment, in which the bits of corneal tissue were first placed in contact with vaccine virus and then with immune serum, was also made. At times the tissues were allowed to remain in contact with immune serum or vaccine virus overnight in the icebox. The results of such experiments have been irregular, and, inasmuch as no definite conclusions can be derived from them, a report concerning them will not be made at this time.

Certain experiments of other sorts, properly performed and controlled, yielded constant results. They were the ones in which small pieces of normal corneas, inoculated with vaccine virus either *in vivo* or *in vitro*, were cultivated in normal and in antivaccinal plasma for 24 and 48 hours, or in which immune corneas were removed from animals, washed for 2 hours in Ringer's or Tyrode's solution, placed

in a virus emulsion for 2 or 3 hours, and finally cultivated for 24 and 48 hours in normal and in immune plasma. It seems unnecessary to give in detail all the experiments. Some of the typical ones, with their results, will be described however.

Development of Vaccinal Lesions in Bits of Normal Corneas Cultivated in Antivaccinal Plasma

In the study of vaccinal lesions that developed in normal corneas cultivated in antivaccinal plasma, 5 experiments, in which the corneas were inoculated *in vivo*, and 7, in which they were infected *in vitro*, were performed. Of the former, in one of which glycerolated virus was used, all resulted successfully. Of the latter, 5 resulted successfully; in one the infection was so mild that no conclusions could be drawn, and in another the cultures were sufficiently contaminated with bacteria to be rendered valueless. 2 experiments in which the corneas were inoculated *in vivo*, and 1 in which they were infected *in vitro* will be described in detail.

The Development in Normal Corneas Infected in Vivo and Cultivated in Normal and in Immune Plasma

Experiment XIV. May 14, 1929.—Freshly prepared Levaditi neurovaccine was used. Immune plasma, known to possess sufficient neutralizing power, was collected. Both eyes of 2 normal rabbits were thoroughly cocaineized, scarified, and inoculated with vaccine virus. 2 hours later normal plasma was obtained from the animals. Then the rabbits were sacrificed, the eyes were enucleated, and finally the spleens were removed for tissue extract. The eyes were washed in Ringer's solution in order to remove as many contaminating bacteria as was possible. The 4 corneas were removed and divided into 20 pieces, 4 of which were immediately placed as controls in Zenker's fluid. The remaining 16 were treated in the following manner:

A.—In clots of normal plasma and spleen extract, 4 pieces of cornea were cultivated at 37°C. for 24 hours, after which 2 were fixed in Zenker's fluid and examined for vaccinal lesions, and 2 were carefully removed from the plasma medium, gently washed in Ringer's solution, ground in a mortar, and injected into the skin of a normal rabbit as a test for the presence of active vaccine virus.

B.—4 pieces were treated in a manner similar to that employed under A with the exception that immune plasma was used instead of normal.

C and D.—Duplicates of A and B respectively, with the exception that the cultures were incubated 48 hours instead of 24.

The results of the above experiment are summarized in Table I and clearly show that normal corneas, inoculated with vaccine virus *in vivo* (2 hours) and then cultivated *in vitro* in antivaccinal plasma, developed typical lesions with Guarnieri bodies in the presence of antibodies in the immune plasma. Furthermore, when bits of normal corneas, inoculated and cultivated in a similar manner, were removed from the immune plasma, washed in Ringer's solution,

TABLE I
Summary of Experiments XIV, XVI, and XVII

Exp.	Cultures	24 hours			48 hours		
		Sections	Rabbit inoculations		Sections	Rabbit inoculations	
			Washed	Unwashed		Washed	Unwashed
XIV	N.C. + N.P.	+++	++++	0	++++	++++	0
	N.C. + I.P.	++	++++	0	+++	+++	0
XVI	N.C. + N.P.	?	0	—	+	0	+++
	N.C. + I.P.	?	—	—	+	++	?
XVII	N.C. + N.P.	++	0	+++	+++	0	+++
	N.C. + I.P.	+++	+	±	++	+	+

N = normal; C = cornea; I = immune; P = plasma; 0 indicates that test was not made; ? indicates doubtful lesions or doubt regarding presence of active virus; — indicates absence of lesions or absence of active virus; + s indicate the severity of the lesions with Guarnieri bodies as seen in sections of the bits of cornea, or the amount of active virus as determined by the size of the lesions at the sites of inoculation of the bits of cornea in the skin of rabbits.

emulsified, and injected into the skin of a normal rabbit, an abundance of active vaccine virus was demonstrated. The lesions in the bits of cornea cultivated in immune plasma were not as extensive as were those in the pieces of tissue set up in normal plasma. Nevertheless, the lesions were definite and frequently involved the new cells growing in to repair the defects caused by scarification. Inasmuch as these young cells were formed after the bits of cornea were placed in immune plasma,* the indications are that new cells arising from old infected

* This is indicated by the fact that in the controls, promptly fixed in Zenker's fluid after the inoculated eyes were enucleated, no growth of cells had taken place and no Guarnieri bodies were present.

cells or in the immediate vicinity of involved cells are capable of becoming infected even when the tissues are growing in a medium of immune plasma.

Experiment XVI. June 4, 1929.—The inoculations, collections of normal and immune plasmas, and preparations of cultures, were similar to those described in Exp. XIV. The following tests and controls, subsequently handled as indicated, were set up.

1 bit of inoculated normal cornea in normal plasma and spleen extract incubated 24 hours; examined histologically.

1 bit of inoculated normal cornea in normal plasma and spleen extract incubated 24 hours; ground up with at least 1 cc. of the plasma and injected into the skin of a rabbit.

2 bits of inoculated normal cornea in immune plasma and spleen extract incubated 2 hours; examined histologically.

2 bits of inoculated normal cornea in immune plasma and spleen extract incubated 24 hours; removed from the plasma clots, washed in Ringer's solution, emulsified, and injected into the skin of a normal rabbit.

2 bits of inoculated normal cornea in immune plasma and spleen extract incubated 24 hours; ground up with at least 1 cc. of the plasma and injected into the skin of a rabbit.

1 tube of immune plasma and spleen extract incubated at 37°C. for 24 hours. Then the plasma was tested for its neutralizing power.

1 tube of immune plasma and spleen extract placed in the icebox for 24 hours. Then the plasma was tested for its neutralizing power.

Duplicates of the above preparations were set up and incubated at 37°C. or stored on ice for 48 hours instead of 24.

The results of Experiment XVI are shown in Table I. The control portions of immune plasma held in the icebox and in the incubator for 24 and 48 hours were found to be more than capable of neutralizing the amount of virus in the bits of cornea at the beginning of the experiment. It seemed, however, that both of the 48-hour specimens had lost some of their neutralizing properties. This phenomenon is undergoing further investigation. There was very little evidence of vaccine virus either in the sections of the 24-hour cultures or in the rabbits inoculated with emulsions of the cultures. This was equally true of preparations of normal cornea in normal or in immune plasma. The demonstration of virus in the 48-hour cultures of normal cornea in immune plasma was accomplished through the removal, by gentle washing, of the immune plasma from the bits of tissue before they were

emulsified for test inoculations in rabbits. The failure to separate the infected tissues from the immune plasma may account for the fact that other workers have been unable to demonstrate active virus in cultures of normal tissues infected and then grown in immune plasma. At times,—this fact is well exemplified by the next experiment,—it was possible to demonstrate virus in such cultures even when the pieces of cornea had not been entirely freed from the antivaccinal plasma.

The Development in Normal Corneas Infected in Vitro and Cultivated in Normal and in Immune Plasma

Experiment XVII. June 11, 1929.—This experiment was a repetition of Experiment XVI with 2 exceptions, viz., the corneas were inoculated *in vitro* instead of *in vivo*, and 2 pieces of the tissue were chosen for controls after they had been inoculated with vaccine virus and before the cultures were set up. 1 of the control bits of cornea was ground up with 0.5 cc. of normal plasma, the other with 0.5 cc. of immune plasma. These emulsions, tested in the skin of a normal rabbit, showed that the immune plasma neutralized the virus in the freshly inoculated cornea while the normal plasma did not. Thus it was demonstrated that 3 cc. of immune plasma, the amount used in the cultures, was at least 6 times greater than the amount necessary to neutralize the virus in the bits of cornea at the time when the cultures were set up.

From the results of Experiment XVII, which are summarized in Table I, it is evident that normal corneas infected *in vitro* and cultivated in antivaccinal plasma developed typical vaccinal lesions with Guarnieri bodies and that such cultures contained active vaccine virus. There was little or no difference in the size of the lesions produced by the washed and unwashed bits of cornea from the 48-hour cultures. A difference was apparent, however, in the tests made with the tissue from the 24-hour cultures.

Reaction of Immune Corneas to Vaccine Virus

In the investigations concerning the nature of the *in vitro* reaction of immune corneas to vaccine virus, 6 experiments were performed. 2 of these must be disregarded, inasmuch as the corneas were not tested *in vivo* for immunity before being used in the cultures. For immunity to vaccine virus is relative, and certain tissues more slowly

attain a completely refractory state than do others. This is particularly true of corneal epithelium. In fact, one cornea of an animal may be refractory while the other is slightly susceptible. Moreover, the results of skin tests are not reliable indicators of the state of corneal immunity. Therefore, in order to be correctly informed concerning the state of immunity of any cornea, one must test it directly with a freshly prepared potent virus. In 3 of the 4 experiments which were properly controlled, the immune corneas exhibited no typical vaccinal lesions, while in 1 the corneas showed evidences of a very mild infection associated with a few Guarnieri bodies. 2 of the experiments will be described in detail.

Immune Corneas Infected in Vitro and Cultivated in Normal and in Immune Plasma

Experiment XI. Jan. 18, 1929.—A normal rabbit received multiple intradermal inoculations of vaccine virus. Feb. 8, 1929, the animal was reinoculated in the skin and corneas. A mild reaction occurred in the skin, and a moderately severe one in the eyes. April 16, 1929, the rabbit was sacrificed for immune plasma and immune corneas. A normal animal was likewise sacrificed for plasma, corneas, and spleen extract. A fresh testicular virus emulsion was prepared. The immune and normal corneas were inoculated with vaccine virus *in vitro* (3 hours) in the usual manner; cultures were prepared and incubated at 37°C. for 48 hours; the tissues were then fixed and examined with the following results.

1.—Normal corneas infected and cultivated in normal plasma revealed severe vaccinal lesions with numerous Guarnieri bodies.

2.—Immune corneas infected and cultivated in normal plasma evidenced a very mild infection associated with a moderate number of vaccine bodies.

3.—Immune corneas infected and cultivated in immune plasma showed an exceedingly mild infection with only an occasional Guarnieri body.

In the above experiment, although evidences of a mild infection were found in the immune corneas, a definite difference in the extent of the involvement of the normal and immune tissues was evident. The experiment was repeated with an animal that had been immunized and tested in a manner similar to that used upon the immune animal above, and no evidences of a vaccinal infection were found in the immune corneas, while moderately severe lesions were noted in the normal corneas. In these experiments, no tests for the presence of

active vaccine virus in the cultures were made. Such tests, however were conducted on 2 other occasions, and the results, one set of which will be described below, were identical in both instances.

Experiment XVIII. Three rabbits received multiple intradermal inoculations of vaccine virus, Jan. 24, Feb. 4, and Feb. 18, 1929, respectively. June 3, 1929, the animals were reinoculated in the skin and corneas. All were refractory to reinfection of the skin. The first animal, however, had a mild "take" in the right eye, none in the left; the second evidenced a mild infection in both eyes; the third showed a reaction in the right eye, none in the left. The lesions in the eyes rapidly healed, and the corneas were used as immune tissues. The corneas that had evidenced "takes" and had recovered were labeled A, while those that had shown no reaction were marked B. A's and B's were run in parallel series, but, inasmuch as the results in the 2 series were identical, they will not be described separately.

June 18, 1929, the 3 immune rabbits were sacrificed, and the corneas were removed, scarified, divided into 4 pieces each, and washed for 2 hours in Tyrode's solution at 37°C. Then, the bits of cornea were immersed for 2 hours in a fresh virus emulsion at 37°C. Immune plasma was collected from rabbits recently (2 to 4 weeks) recovered from severe vaccinal infections in the skin. Normal plasma, normal corneas, and spleen extract were obtained from stock rabbits. The normal corneas were scarified and infected in a manner similar to that employed for the immune tissues. Either 4 or 8 tubes of each combination of the materials enumerated above were prepared. One half of the preparations was incubated 24 hours, while the remaining half was held at 37°C. for 48 hours. At the end of each period of incubation the preparations were again divided; the bits of cornea in one half were fixed and stained for histological examination, while those in the other half were carefully removed from their plasma clots, washed in Ringer's solution, emulsified, and injected into the skin of normal rabbits. Lesions produced by these intradermal inoculations were excised, emulsified, and rubbed on the scarified skin of another set of normal rabbits in order to demonstrate typical vaccinal pustules. Control specimens of immune plasma held on ice or in the incubator for 24 and 48 hours were tested for neutralizing properties. The results of the experiment are given in detail below.

24-hour Preparations

- 1.—Normal cornea in normal plasma: Mild vaccinal lesions found, virus present.
- 2.—Immune cornea in normal plasma: No vaccinal lesions observed, virus present.
- 3.—Immune cornea in immune plasma: No vaccinal lesions observed. Virus not demonstrable.
- 4.—Specimens of immune plasma held on ice and in the incubator appeared to have retained full neutralizing power.

48-hour Preparations

5.—Normal cornea in normal plasma: Pronounced vaccinal lesions observed, large amount of virus present.

6.—Immune cornea in normal plasma: No vaccinal lesions found, virus present but less in amount than that in tissue of 5.

7.—Immune cornea in immune plasma: No vaccinal lesions seen, virus present but less in amount than that found in the cultures of 6.

8.—Specimens of immune plasma held on ice and in the incubator seemed to have lost some of their neutralizing power. They were still capable, however, of neutralizing an amount of virus greater than that taken up by the bits of cornea during the process of inoculation.

From the results of Experiment XVIII it appears that the epithelium of normal and immune corneas did not react similarly to vaccine virus, as evidenced by the fact that typical vaccinal lesions with Guarneri bodies were observed only in the former. Virus, however, was demonstrated in the preparations of normal corneas in normal plasma, in the 24-hour cultures of immune corneas in normal plasma, and in the 48-hour cultures of immune corneas in normal and in immune plasma. Furthermore, the amount of virus in the 48-hour preparations of immune corneas seemed, as determined by the size of lesions produced in rabbits, to be greater than that in the 24-hour cultures. At present, it is impossible to state whether this apparent increase in the amount of virus was due to an actual multiplication of the infectious agent. Nevertheless, the striking fact remains that vaccine virus can at least survive for 48 hours in cultures of immune corneas in normal plasma and to a less extent in preparations of immune corneas in antivaccinal plasma.

DISCUSSION

It should be noted that the *in vitro* vaccinal infection of normal corneal epithelial cells and the *in vitro* refractory state exhibited by immune corneal epithelium occurred without the direct intervention of such structures as cells of the reticulo-endothelial system and white blood cells. What part, if any, the various tissues of the body played in bringing about the refractory state of the immune corneas before they were removed from the body is not known.

Adequate explanations of some of the observations reported in this

paper must await further experimental work. Interpretations for a few of the results, however, are at hand, but even these are associated with a certain amount of speculation. The fact that vaccinal lesions with Guarnieri bodies developed in normal corneas inoculated *in vitro* and then cultivated in antivaccinal plasma does not appear unusual, provided one assumes that the activity of virus situated intracellularly could not be influenced by the antibodies in the immune plasma. This assumption at least agrees with the experimental observations of Rous and Jones (24) who found that intracellularly situated typhoid bacilli and red blood cells were not susceptible to such injurious agents as potassium cyanide and antisera. The results of the experiments in which immune corneas* were cultivated in normal and in immune plasma appear to support the idea of Todd (17) and Andrewes (18) that substances protective against viruses are not necessarily virucidal and also to fall in line with the observations recorded concerning the persistence of viruses in immune animals (20, 21, 22, 23).

SUMMARY

Normal corneas inoculated *in vitro* with vaccine virus and then cultivated in antivaccinal plasma developed typical vaccinal lesions associated with Guarnieri bodies. In such cultures, after an incubation period of 24 or 48 hours, active vaccine virus was demonstrated by means of appropriate methods.

Immune corneas inoculated *in vitro* with vaccine virus and then cultivated in normal or in antivaccinal plasma revealed either very mild vaccinal lesions or none at all. In some of the cultures after 24 and 48 hours of incubation, active vaccine virus was demonstrated.

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* Results reported in this paper are different in certain respects from those secured by Dr. Andrewes (personal communication to Dr. Rivers). Such differences, no doubt, may be accounted for by the fact that he conducted his experiments with Virus III in testicular tissue cultures made according to Maitland's technique.

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